

Domingo Ranieri

Address: Via dell'Indipendenza 67/2, Bologna, Italy

E-mail: domiran98@gmail.com

Date of birth: 24/01/1998 * *Citizenship :* Italian, Australian

Profile

I hold a Master's degree in Theoretical Physics, where I built a strong foundation in Python programming, numerical methods, and data analysis using tools such as Pandas, NumPy, Matplotlib, TensorFlow, Keras, and PyTorch. My studies also strengthened my problem-solving abilities and my capacity to model complex systems, skills that I now apply across advanced data engineering, AI, and distributed computing projects. My experience spans machine learning engineering, blockchain-enhanced data management, federated cloud orchestration, quantum computing, and modern data lakehouse architectures.

Education

Master's Degree in Physics

Theoretical Physics

Final grade: 110/110 cum laude

Thesis: Simulation of neuromuscular control using a quantum computer

University of Bologna

2020 - 2022

Bachelor's Degree in Physics

Physics

Final grade: 110/110 cum laude

Thesis: Fractal Universe model and Cosmic Acceleration

University of Bologna

2017 - 2020

Work experience

ICSC

Software Engineer

Mar 2025 - Present

Bologna, Italy

- Research and development in AI, data engineering, and quantum computing
- Design and implementation of advanced software services
- Delivery of technical courses on AI, cloud, and data engineering
- Participation in conferences and collaborative research projects

ICSC-INFN

Research Fellow

Sep 2023 - Feb 2025

Bologna, Italy

- Development of AI and blockchain-based services
- Applied research in distributed and intelligent systems
- Participation in workshops and scientific events

Key Projects

INFN Data Cloud

Federated Cloud Orchestration & AI-Ranker

2024 - 2025

- Designed and contributed to the architecture of a federated cloud orchestration platform (Indigo-Orchestrator) enabling automated deployment across distributed cloud providers
- Developed the AI-Ranker component to optimize deployment decisions based on dynamic metrics
- Built and managed machine learning pipelines using **scikit-learn** and **MLflow** (experiment tracking, model versioning, deployment)

- Implemented real-time data ingestion pipelines with **Apache Kafka** and feature engineering workflows with **Pandas** and **NumPy**
- Delivered end-to-end integration from data ingestion to model inference, improving system scalability, automation, and decision accuracy

Domain-Specific RAG System

Genetic Research

2026 - Present

- Designed a **domain-adapted Retrieval-Augmented Generation (RAG)** system for querying and synthesizing biomedical literature
- Implemented hybrid retrieval strategies combining **vector search** and **GraphRAG**
- Built scalable multi-source data pipelines using **Milvus** for vector storage
- Developed an **agentic RAG architecture** capable of adapting retrieval strategies based on query intent (exploratory vs evidence-based)
- Improved traceability and reliability of generated outputs through evidence-grounded responses and source attribution

Multi-Agent Chatbot

Genetic Pathogenicity Analysis

2026 - Present

- Architected a **multi-agent AI system** for genetic variant interpretation and pathogenicity analysis
- Designed modular agent workflows with task delegation and reasoning pipelines using **CrewAi**
- Integrated **Model Context Protocol (MCP)** for standardized tool interaction and context sharing across agents
- Developed custom tools for real-time data analysis, including API integration, automated workflows, and scientific data processing
- Enabled generation of analytical outputs (charts, reports) for decision support and research workflows

Technical skills

Programming	Python, C++, Go, Bash
ML/AI	TensorFlow, PyTorch, Keras, Scikit-learn
Data Engineering	Kafka, Airflow, Iceberg, Trino, MinIO
Cloud & DevOps	Docker, Kubernetes, Ansible
Databases	PostgreSQL, MongoDB, MySQL
Web	Flask, FastAPI, HTML, CSS, JavaScript
Tools	MLflow, OpenMetadata, Great Expectations, Git

Languages

Italian	Native
English	B2

Communication and interpersonal skills

- Effective presentation and public speaking
- Strong technical writing and documentation skills
- Collaborative teamwork in multidisciplinary environments
- Ability to explain complex concepts clearly